

METAL SOLUTIONS

EOS StainlessSteel CX

Material Data Sheet

EOS STAINLESSSTEEL CX

Combines Corrosion Resistance with High Strength and Hardness

EOS StainlessSteel CX is a tooling grade steel characterized by having a good corrosion resistance combined with high strength and hardness. Parts built from EOS StainlessSteel CX can be machined, shot-peened and polished in as manufactured or heat treated state.

MAIN CHARACTERISTICS

- Stainless steel with excellent corrosion resistance combined with high strength and hardness
- The parts are easily machinable and offer excellent polishability
- The parts offer excellent wear and fatigue resistance

TYPICAL APPLICATIONS

- Plastic injection molding tools and tool parts for demanding applications
- Rubber molding tools and tool parts
- Molding tools and tool parts for corrosive plastics
- Other industrial applications where high strength and hardness are required

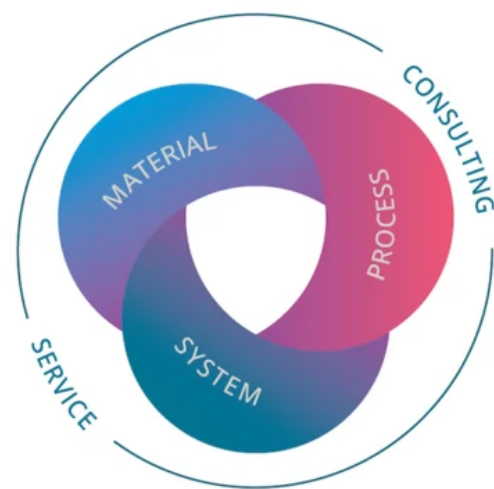
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



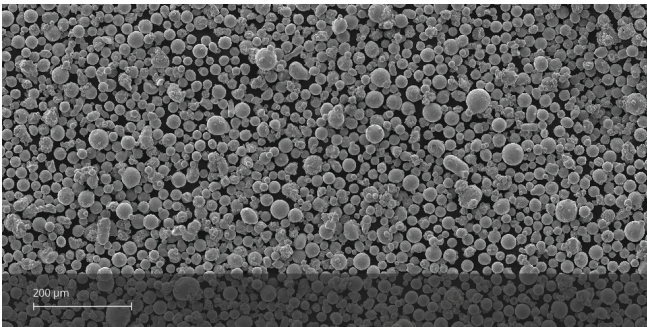
POWDER PROPERTIES

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe		Balance
Cr	11	13
Ni	8.4	10
Mo	1.1	1.7
Al	1.2	2
Mn	0	0.4
Si	0	0.4
C	0	0.05

Powder Particle Size

Generic Particle Size Distribution	20 - 65 µm
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SEM micrograph of EOS StainlessSteel CX powder

HEAT TREATMENT

Description

EOS StainlessSteel CX can be heat treated to match various needs of different applications. The two step heat treatment can be performed under vacuum or inert gas atmosphere. First step is solution annealing to minimize amount of austenite in the martensitic matrix. The needed hardness and strength is achieved through aging treatment where precipitation hardening takes place

Steps

Solution Annealing: 30 minutes at 850 °C (± 10 °C) measured from the part followed by rapid air cooling to room temperature (below 32 °C). Cooling rate 20-60 °C/min. Reaching room temperature before starting aging treatment is required to achieve desired microstructure.

Aging: For peak hardness and strength 2 h at 525 °C (± 10 °C) measured from the part followed by air cooling. Mechanical properties presented in this document achieved through this aging procedure.

EOS StainlessSteel CX for EOS M 290 | 30µm

EOS M 290 - 30 µm - TRL 7

Process Information Metal

This process product is optimized for flexible and reliable production of CX parts with emphasis on quality.

System Setup	EOS M 290
EOS Material set	CX_030_HiPerM291
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	30 µm
Volume Rate	3.2 mm³/s
Typical Dimensional Change after HT [%]	+0.1 %

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
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Average Defect Percentage	-	0.03 %	-
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Density EN ISO 3369	Thickness	Result	Number of Samples
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Average Density	-	≥ 7.69 g/cm³	-
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Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1670	1730	6	-	-	-
Horizontal	1650	1720	7.5	-	-	-

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	770	1050	12	-	-	-
Horizontal	810	1110	16	-	-	-

Hardness

EN ISO 6508 Heat Treated	
Value	50
Unit	HRC

EN ISO 6508 As Manufactured	
Value	33
Unit	HRC

Fatigue

Heat Treated	
Fatigue strength [MPa]	695

Fatigue strength at 1 x 10 million cycles in heat treated state ASTM E466
Fatigue strength determines a stress level where specimen fails at a defined number of stress cycles [ISO 12107]. Fatigue strength was estimated statistically according to ISO 12107. Testing was done according to ASTM E466. Fatigue results typically show large deviations due to the nature of the fatigue process [ISO 12107].

Coefficient of Thermal Expansion

ASTM E228	Temperature
11.1*10 ⁻⁶ /K	25 – 100 °C
11.6*10 ⁻⁶ /K	25 – 200 °C
11.9*10 ⁻⁶ /K	25 – 300 °C
12*10 ⁻⁶ /K	25 – 400 °C

EOS StainlessSteel CX for EOS M 290 | 40/80 µm

EOS M 290 - 40 µm - TRL 3

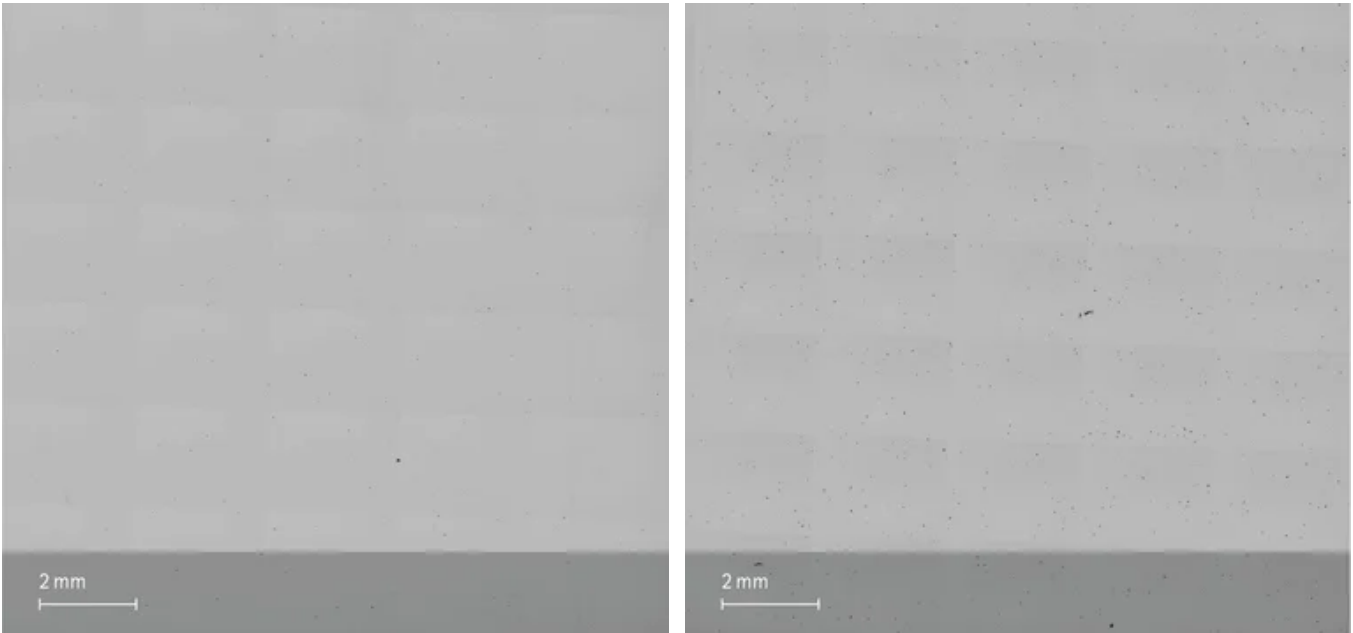
Process Information Metal

This process product is optimized for flexible and fast production of CX parts with EOS M 290 system. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

System Setup	EOS M 290
EOS Material set	CX_040_080_Core M291
Required Software Plan	EOS BUILD
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Nitrogen
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1 -7.8 mm³/s
Typical Dimensional Change after HT [%]	+0.1 %

Chemical and Physical Properties of Parts



Etched micrograph of powder with 40 µm (left), 80 µm (right)

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	<0.1 %	-
Average Defect Percentage	80 µm	<0.2 %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1640	1720	6	-	-	-
Horizontal	1630	1720	7	-	-	-

Mechanical Properties Heat Treated

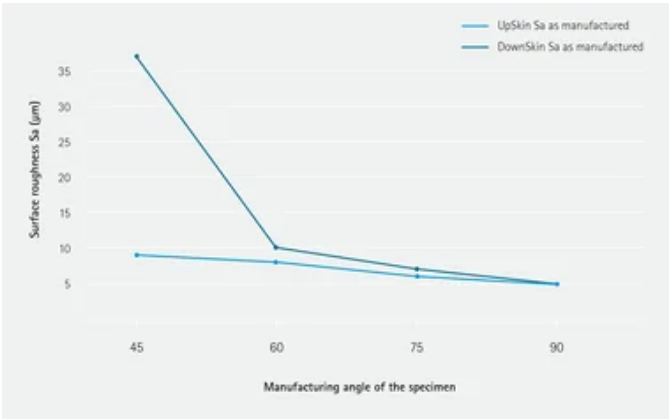
EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1600	1680	5.5	-	-	-
Horizontal	1590	1700	6.5	-	-	-

Impact Toughness

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	14
Layer Thickness	40 µm

Heat Treated	
Typical Impact Toughness [J]	10
Layer Thickness	80 µm

Surface Roughness



EOS StainlessSteel CX for EOS M 400-4 | 40/80 µm

EOS M 400-4 - 40 µm - TRL 3

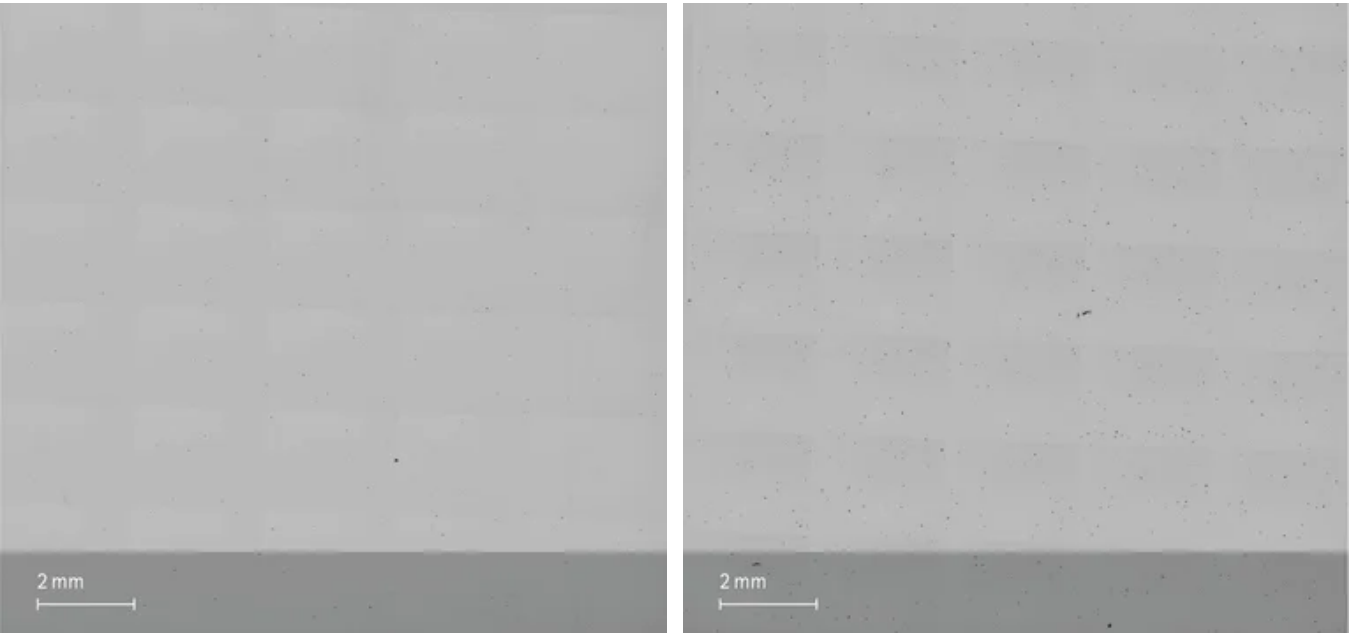
Process Information Metal

This process product is optimized for flexible and fast production of CX parts with EOS M 400-4 system. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

System Setup	EOS M 400-4
EOS Material set	CX_040_080_Core M404
Required Software Plan	EOS BUILD
Software Requirements	EOSPRINT 2.16 or higher EOSYSTEM 2.17 or newer
Recoater Blade	Ceramic
Nozzle	Standard
Inert gas	Nitrogen
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1 -7.8 per laser mm³/s
Typical Dimensional Change after HT [%]	+0.1 %

Chemical and Physical Properties of Parts



Etched micrograph of powder with 40 μm (left), 80 μm (right)

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
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Average Defect Percentage	40 μm	<0.1 %	-
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Density	Thickness	Result	Number of Samples
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Average Density	80 μm	<0.2 g/cm ³	-
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Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1650	1720	5.5	-	-	-
Horizontal	1650	1720	6.5	-	-	-

Mechanical Properties Heat Treated

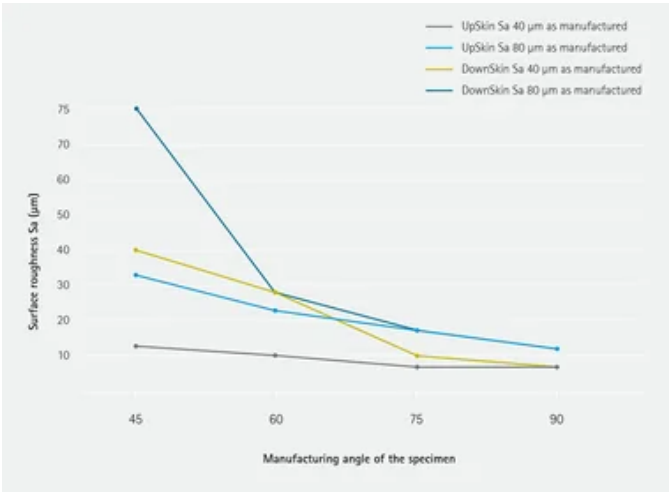
EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1600	1690	4.5	-	-	-
Horizontal	1610	1700	5.5	-	-	-

Impact Toughness

EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	14
Layer Thickness	40 µm

Heat Treated	
Typical Impact Toughness [J]	10
Layer Thickness	80 µm

Surface Roughness



HEADQUARTERS

EOS GmbH
Electro Optical Systems

Robert-Stirling-Ring 1
82152 Krailling / Munich
Germany

Tel.: +49 89 893 36-0
Email: info@eos.info
URL: www.eos.info

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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